

HO CHI MINH UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

STATISTICAL LEARNING



FINAL PROJECT REPORT

**Dearly: Secure Virtual Assistant for Elderly Care
with Speaker Verification and Identification**

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Abstract

Dearly is a Vietnamese Android virtual assistant designed for elderly users and their caregivers. The application accepts recorded speech, transcribes the request, selects an assistant action, and returns both text and a Vietnamese spoken response. Its core security requirement is that a voice request alone must not authorize a sensitive action. Dearly therefore combines automatic speech recognition (ASR), speaker verification (SV), speaker identification (SID), and server-side authorization. A fine-tuned ECAPA-TDNN embedding model supports both one-to-one verification and one-to-many identification [1, 2]; Whisper supports Vietnamese transcription [3].

Contents

Abstract	2
Group Members and Contribution	5
1 Introduction and System Overview	6
1.1 Project Context and Objectives	6
1.2 Required Assistant Use Cases	6
1.3 System Architecture	6
1.4 Voice-Request Processing Flow	7
2 Dataset Preparation and Model Training	8
2.1 VIVOS Dataset and Speaker-Disjoint Splits	8
2.2 Audio Preprocessing and Speaker Enrollment	8
2.3 ECAPA-TDNN Model and Training Procedure	8
3 Experimental Results and System Integration	10
3.1 Evaluation Protocol and Experimental Results	10
3.2 Speaker Enrollment, Verification, and Identification Integration	10
3.3 Implementation Validation and Demonstration Evidence	11
3.4 Limitations and Conclusion	12
References	13

List of Figures

1	Dearly service architecture. The Android app does not call the database, Kafka, or AI service directly.	7
2	Speaker-verification-gated medication action. A first query never authorizes the mutation.	11
3	Reserved evidence: speaker enrollment workflow.	11
4	Reserved evidence: unauthenticated general voice function.	12
5	Reserved evidence: SV-protected medication action.	12
6	Reserved evidence: SID personalization.	12

List of Tables

1	Project group members.	5
2	Task distribution and contribution.	5
3	Dearly voice-assistant use cases and security controls.	6
4	Speaker-disjoint VIVOS split used by the completed experiment.	8
5	Fine-tuning configuration for the VIVOS ECAPA-TDNN experiment.	9
6	Held-out VIVOS evaluation results.	10

Group Members and Contribution

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Table 1: Project group members.

No.	Work package	Responsible member	Contribution
1	Android client, voice-capture and enrollment user experience, caregiver/elder workflows, and integration testing.	Lê Nguyên Thảo	50%
2	ECAPA-TDNN training and evaluation, FastAPI and Go voice services, authorization design, experiment analysis, and report integration.	Đinh Đại Vũ	50%

Table 2: Task distribution and contribution.

1 Introduction and System Overview

1.1 Project Context and Objectives

Older adults benefit from hands-free access to time, medication information, and caregiver-supported routines. At the same time, commands that change health-related state must resist accidental or unauthorized execution. Dearly addresses this tension with the following objectives:

- provide voice-based interaction in Vietnamese through an accessible Android interface;
- execute a general information request without speaker authentication;
- require successful SV before a medication dose is marked as taken; and
- use SID only for personalization, limited to a caller’s trusted caregiver–elder group.

1.2 Required Assistant Use Cases

Table 3 maps the required project behaviors to the implemented assistant functions. The first query is intentionally informational: it may detect that a protected action was requested, but it cannot mutate medication data. A second recording is required for SV before the protected endpoint accepts the action.

Category	Example function	Control
General	Ask for the current time/date or medication schedule.	No SV; response uses local time or stored schedule.
Important	Mark a medication dose as taken.	One-to-one SV and a short-lived, one-use voice grant.
Personalized	Return a response addressed to a recognized linked family member.	One-to-many SID over only the caller and directly linked caregiver/elder profiles.

Table 3: Dearly voice-assistant use cases and security controls.

1.3 System Architecture

Figure 1 shows the production design. Android is the only public client. It records audio, sends it to the Go API, displays the response, and uses Android’s local `vi-VN` TextToSpeech engine for output. The Go API authenticates sessions, manages enrollment and medication state, and calls the private FastAPI service. The AI service runs ASR, ECAPA embedding extraction, SID, SV, and constrained LLM/local intent analysis. PostgreSQL is authoritative

for user, relationship, medication, enrollment, and grant data; Kafka delivers domain events to the notification worker.

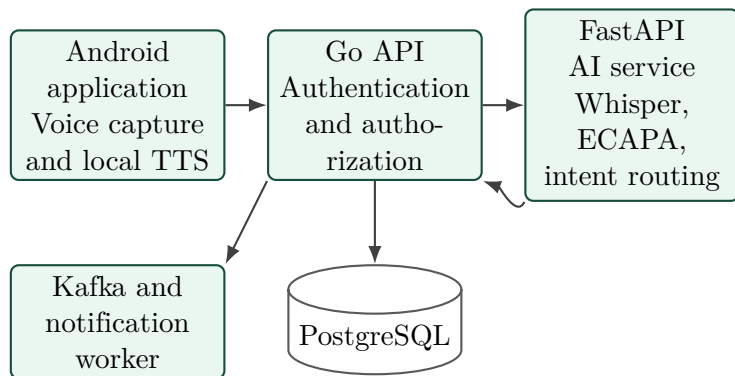


Figure 1: Dearly service architecture. The Android app does not call the database, Kafka, or AI service directly.

1.4 Voice-Request Processing Flow

For a normal query, the AI service transcribes the recording with Whisper, identifies a speaker when trusted profiles are supplied, and classifies the request. The API enriches allowed general intents such as `ASK_TIME`, `ASK_DATE`, and `CHECK_MEDICATIONS` from local time or PostgreSQL. A protected intent returns a response requesting verification instead of performing the action. This separation ensures that ASR and intent classification do not become authorization decisions.

2 Dataset Preparation and Model Training

2.1 VIVOS Dataset and Speaker-Disjoint Splits

The speaker-recognition experiment uses VIVOS, a Vietnamese speech corpus released for academic use under the CC BY-NC-SA 4.0 license [4]. VIVOS is appropriate for this project because the final assistant operates in Vietnamese. The completed run places every speaker in exactly one partition, avoiding identity leakage between training, checkpoint selection, and final evaluation.

Split	Speakers	Utterances	Purpose
Training	41	10,310	Fine-tune the ECAPA embedding model and AAM-Softmax head.
Validation	5	1,350	Select the checkpoint with the best verification validation loss.
Test	19	760	Hold-out speaker-verification and speaker-identification evaluation only.

Table 4: Speaker-disjoint VIVOS split used by the completed experiment.

2.2 Audio Preprocessing and Speaker Enrollment

The training pipeline builds manifests containing utterance identifiers, absolute audio paths, speaker identifiers, and durations. At load time, recordings are converted to mono, resampled to 16 kHz when necessary, and transformed into fixed three-second examples. Training crops use a random start point; validation uses a deterministic center crop. Short recordings are zero-padded. This makes batches comparable while preserving the original speaker-disjoint assignment.

The Android enrollment workflow requests five prompted phrases. Each phrase produces a 192-dimensional ECAPA embedding. The backend stores the individual embeddings as serialized `float32` vectors and derives an average profile for verification. The workflow does not persist raw enrollment audio after the embedding is obtained. The five phrase records also provide the profile material used in the held-out SID protocol.

2.3 ECAPA-TDNN Model and Training Procedure

ECAPA-TDNN extends TDNN-based speaker representation learning with channel attention, multi-layer feature aggregation, and attentive statistics pooling [1]. Dearly starts from the SpeechBrain pretrained `speechbrain/spkrec-ecapa-voxceleb` source and fine-tunes the embedding model on VIVOS [2]. The deployed runtime produces a finite 192-dimensional embedding, compares profiles by cosine similarity, and applies separate configurable SV and SID thresholds.

Setting	Completed experiment value
Compute environment	Google Colab Tesla T4 GPU
Final training duration	20 epochs, initialized from the original pretrained source
Batch size and audio input	32 examples; three-second mono 16 kHz crops
Objective	AAM-Softmax, margin 0.2 and scale 30
Optimizer and schedule	AdamW; cosine learning-rate decay from 10^{-4} to 10^{-6}
Backbone schedule	Frozen in epochs 1–5 and unfrozen in epochs 6–20
Checkpoint selection	Minimum validation loss: 0.2644617259502411

Table 5: Fine-tuning configuration for the VIVOS ECAPA-TDNN experiment.

The experimental sequence first evaluated the untouched VoxCeleb-pretrained baseline, then ran a one-epoch smoke experiment, and finally ran the independent 20-epoch training procedure. The final checkpoint is loaded by the production ECAPA wrapper rather than by a separate evaluation-only implementation. This checks that the artifact evaluated for the report can generate the same type of runtime embedding used for SV and SID.

3 Experimental Results and System Integration

3.1 Evaluation Protocol and Experimental Results

Speaker verification is evaluated with 10,000 deterministic balanced trials on held-out VIVOS speakers: 5,000 genuine trials and 5,000 impostor trials. Given two embedding vectors a and b , the runtime score is cosine similarity:

$$\text{score}(a, b) = \frac{a \cdot b}{\|a\|_2 \|b\|_2}.$$

The report uses equal error rate (EER), the point where false reject and false accept rates are equal, and minimum detection cost function (minDCF). SID Top-1 accuracy is computed by averaging five enrollment embeddings for each test speaker, then assigning each remaining query recording to the highest-scoring profile.

System	EER	minDCF	SID Top-1
Untouched VoxCeleb ECAPA baseline	4.40%	0.002978	100.00%
One-epoch smoke model	1.58%	0.001570	100.00%
VIVOS fine-tuned ECAPA	1.28%	0.001628	100.00%

Table 6: Held-out VIVOS evaluation results.

Fine-tuning lowers EER by 3.12 percentage points, or 70.9% relative to the baseline. The experiment’s EER operating threshold is 0.34559793992808. It is not automatically deployed: the current application threshold remains 0.80 and requires calibration on representative elderly-user recordings before it can be changed. The test-set result therefore demonstrates Vietnamese-domain performance, not a clinical or elderly-population validation.

3.2 Speaker Enrollment, Verification, and Identification Integration

Enrollment is available in Android settings. The user reads five prompts; the client captures each recording and posts it with its phrase index to the backend. The backend forwards the audio to the FastAPI enrollment route, receives an ECAPA embedding, upserts the phrase record, and updates the profile average. A query uses stored profiles only from the authenticated caller and directly linked caregiver/elder accounts. The backend checks that an AI-returned identity belongs to that trusted set before exposing `recognized_user_name` for personalization.

Figure 2 describes the protected medication flow. The voice grant is random, valid for two minutes, bound to a user and intent, usable once, and stored as a SHA-256 hash. The medication endpoint consumes the grant transactionally; a request without a valid grant is rejected.

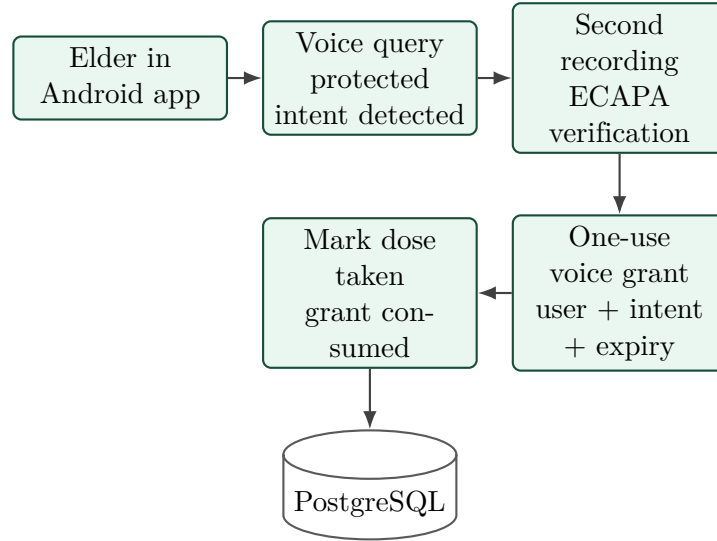


Figure 2: Speaker-verification-gated medication action. A first query never authorizes the mutation.

3.3 Implementation Validation and Demonstration Evidence

The completed engineering checks include AI-service linting and focused tests, Docker Compose configuration validation, a ready AI container with the trained checkpoint, backend unit tests for voice grants and general responses, and a successful Android debug build. These are implementation checks; they do not replace a physical-device demonstration. In particular, the current evidence does not claim successful Firebase authentication, microphone capture, Vietnamese TTS playback, real enrollment, SV, or SID on a device.

The following captioned placeholders reserve the exact visual evidence required for the final submission. They must be replaced with real screenshots or video frames before the report is submitted.

Pending live demonstration evidence

Android settings showing the five-phrase enrollment dialog and a completed 5/5 state.

Replace this framed placeholder with the final device screenshot before submission.

Figure 3: Reserved evidence: speaker enrollment workflow.

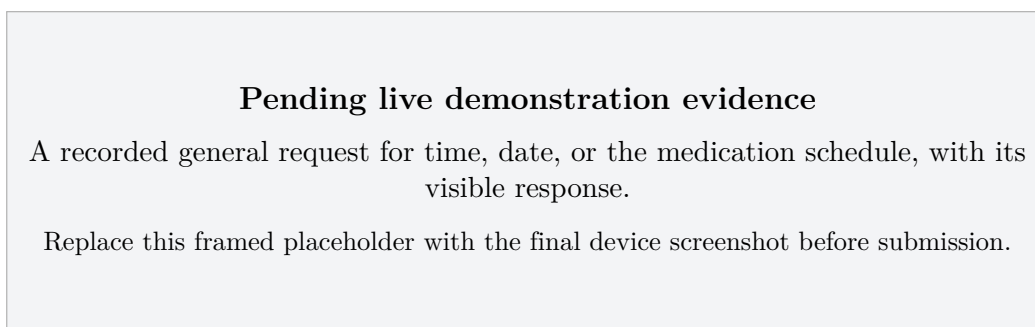


Figure 4: Reserved evidence: unauthenticated general voice function.

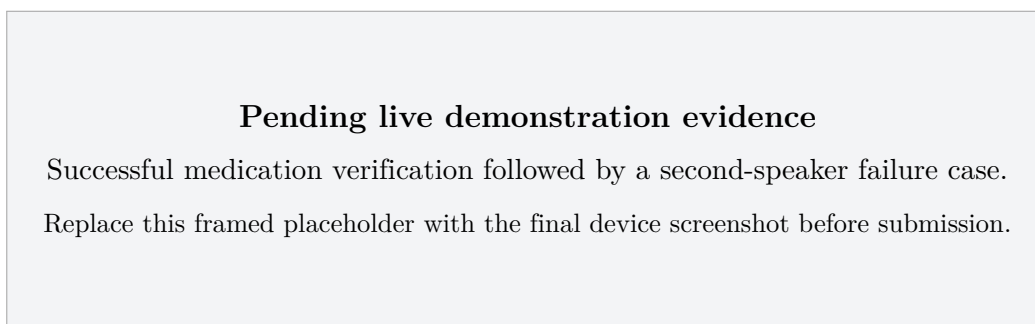


Figure 5: Reserved evidence: SV-protected medication action.

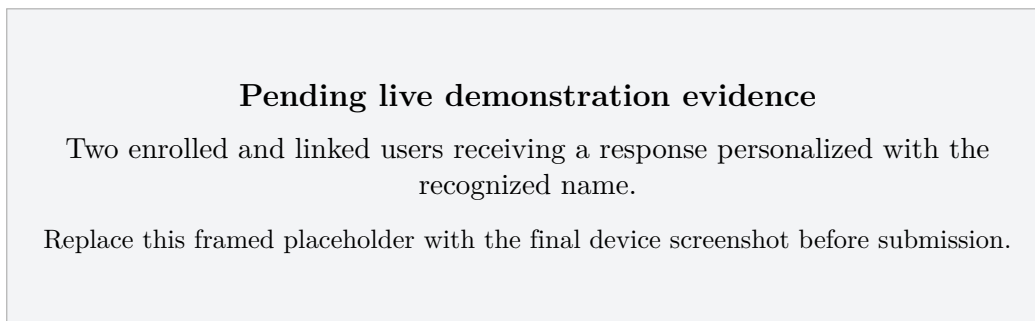


Figure 6: Reserved evidence: SID personalization.

3.4 Limitations and Conclusion

Dearly provides a reportable end-to-end design: VIVOS fine-tuning improves held-out verification performance, ECAPA embeddings drive both SV and SID, and a server-side grant prevents a classified protected voice command from becoming an authorization decision. The remaining limitation is evidence collection rather than a claim of completed live validation. The final package must include real-device demonstration media, source code, the report, and a Drive-link text file for large dataset/model artifacts when needed. The ZIP filename for this team should use both student IDs: `23127118_23127144.zip`.

References

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